Fan Liu

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EDUCATION

Massachusetts Institute of Technology *Cambridge, MA, USA*

- Ph.D. in Mechanical Engineering Sep 2019 – May 2024
  - Thesis: Intercellular flow-mediated force relaxation measurement on the three-dimensional multicellular tissue
  - Adviser: Prof. Ming Guo

Tsinghua University *Beijing, China*

- M.S. in Mechanical Engineering Sep 2016 – Jul 2019

Beihang University *Beijing, China*

- B.S. in Mechanical Engineering (top 1% graduate) Sep 2012 – Jul 2016

RESEARCH APPOINTMENTS & EXPERIENCE

Massachusetts Institute of Technology *Cambridge, MA, USA*

- **Postdoctoral Associate**, Mechanical Engineering Jan 2025 – Present
  - Mentors: Prof. Ming Guo and Prof. Roger D. Kamm
  - Developing mechanics–transport assays to quantify brain tissue poroelasticity in aging and Alzheimer's disease.
  - Investigating how vimentin regulates spheroid poroelasticity and traction-mediated invasion.

- **Doctoral Research**, Mechanical Engineering Sep 2019 – May 2024
  - Developed a parallel-plate micromechanics platform and combined experiments with finite-element modeling to establish how intercellular fluid flow governs tissue force relaxation.
  - Developed mechanically driven bulk electroporation for intracellular delivery.

Tsinghua University *Beijing, China*

- **Master's Research**, Mechanical Engineering Sep 2016 – Jul 2019
  - Advisors: Prof. Jia Cheng and Prof. Lihong Ji
  - Developed triboelectric high-voltage platforms for microplasma generation and electro-assisted cell printing.

PUBLICATIONS

Journal Articles

[13] **Liu F.**, Gao B., Lei L.R., Liu S.N., Li H., and Guo M., "Intercellular flow dominates the poroelasticity of multicellular tissues." *Nature Physics* 21(8), 1311–1318 (2025). **MIT News**, **MIT Spotlight**

[12] Du Y., Wang Y.R., Bao Q.Y., Xu X.X., Xu C., Wang S., Liu Q., **Liu F.**, Zeng Y.L., et al., "Personalized Vascularized Tumor Organoid-on-a-Chip for Tumor Metastasis and Therapeutic Targeting Assessment." *Advanced Materials* 37(6), 2412815 (2025).

[11] Du Y., Xu X.-X., Yu S.-X., Wang Y.-R., Liu Y., **Liu F.**, Liu W., et al., "Dynamics of endothelial cells migration in nature-mimicking blood vessels." *Talanta* 277, 126415 (2024).

[10] **Liu F.**, Yang Z., Yao R., Li H., Cheng J. and Guo M., "Bulk Electroporation for Intracellular Delivery Directly Driven by Mechanical Stimulus." *ACS Nano* 16(11), 19363–19372 (2022).

[9] Yang Z.*, Yang Y.*, **Liu F.***, Li B., Li Y., Liu X. et al., "Thousandfold boosting instantaneous current of triboelectric nanogenerator based on decoupled charge pump and discharge tube." *Nano Energy* 98, 107264 (2022).

[8] Yang Z., Yang Y., Wang H., **Liu F.**, Lu Y., Ji L., Wang Z.L., and Cheng J., "Charge Pumping for Sliding-mode Triboelectric Nanogenerator with Voltage Stabilization and Boosted Current." *Advanced Energy Materials* 11(28), 2101147 (2021).

[7] Yang Z., Yang Y., **Liu F.**, Wang Z., Li Y., Qiu J. et al., "Power backpack for energy harvesting and reduced load impact." *ACS Nano* 15(2), 2611–2623 (2021).

[6] Wang Z., Shi Y., **Liu F.**, Wang H., Liu X., Sun R., et al., "Distributed mobile ultraviolet light sources driven by ambient mechanical stimuli." *Nano Energy* 74, 104910 (2020).

[5] Huo H.*, **Liu F.***, Luo Y., Gu Q., Liu Y. et al., "Triboelectric nanogenerators for electro-assisted cell printing." *Nano Energy* 67, 104150 (2020).

[4] **Liu F.**, Liu Y., Lu Y., Wang Z., Shi Y. et al., "Electrical analysis of triboelectric nanogenerator for high voltage applications exampled by DBD microplasma." *Nano Energy* 56, 482–493 (2019).

[3] Cheng J., Ding W., Zi Y., Lu Y., Ji L., **Liu F.** et al., "Triboelectric microplasma powered by mechanical stimuli." *Nature Communications* 9, 3733 (2018).

[2] Huo H., Cheng J., Lu Y., and **Liu F.**, "Characteristics of Voltage Multiplier Circuits Driven by Triboelectric Nanogenerators." *Mechanical Engineering and Technology* 7(3), 223–233 (2018). (In Chinese.)

[1] Haris M., Watanabe T., **Fan L.**†, Widyanto M.R., and Nobuhara H., "Superresolution for UAV Images via Adaptive Multiple Sparse Representation and Its Application to 3-D Reconstruction." *IEEE Transactions on Geoscience and Remote Sensing* 55(7), 4047–4058 (2017).

(* Equal contribution; †published as Liu Fan.)

Conference Proceedings

[C1] **Liu F.** and Guo M., "Mechanics of multicellular living systems." *Proceedings of SPIE* 13853, *Optical Elastography and Tissue Biomechanics XIII*, 1385304 (2026).

Patents & Patent Applications

[P2] Li H. and **Liu F.**, "Microscale Biological Force Measurement Platform and Method of Use." Chinese patent application **CN120668504A** (application no. 202410313368.7). Published September 19, 2025. *Patent pending.*

[P1] Cheng J., **Liu F.**, Huo H., Zhang Y., Ji L., and Lu Y., "Weight-Reduction Training Rehabilitation Platform." Chinese invention patent **CN107617190B** (ZL201711049698.6). Granted November 15, 2022.

Manuscript in Submission

- **Liu F.**, Rodriguez C.,van Zyl J., et al., Kamm D. R., and Guo M., "Vimentin Regulates Spheroid Poroelasticity to Enable Traction-Mediated Invasion." *In submission.*

AWARDS & FELLOWSHIPS

- Rohsenow Fellowship, Massachusetts Institute of Technology — Jan 2021
- School of Engineering SMA Fellowship, Massachusetts Institute of Technology — Sep 2019
- Outstanding Graduate, Beijing Municipality and Tsinghua University — 2019
- Asian Youth Cultural Ambassador, Tsinghua University — Oct 2018
- National Scholarship for Excellent Academic Performance (2%) — 2013, 2014

PENDING GRANT APPLICATIONS

- NIH/NIA K99/R00 Pathway to Independence Award application
  Principal Investigator; submitted October 2026; pending peer review.

INVITED TALKS & SELECTED CONFERENCE PRESENTATIONS

Invited Talks

- "Vimentin: A Cellular Filament That Softens Cell Aggregates and Promotes Invasion."
  Invited internal seminar, MIT MechE iMMEC Seminar Series, Cambridge, MA, October 6, 2026.
- "Fluid Transport and Mechanics in Living Tissues: From Physical Principles to Disease."
  Invited seminar, School of Mechanical and Aerospace Engineering, Nanyang Technological University, Singapore (virtual), September 7, 2026.
- "Mechanics of multicellular living systems."
  Invited oral presentation, SPIE Photonics West, Optical Elastography and Tissue Biomechanics XIII, San Francisco, CA, January 17, 2026.
- "Intercellular flow-mediated force relaxation measurement in three-dimensional multicellular tissues."
  Invited seminar, School of Systems Science, Beijing Normal University, Beijing, China, November 14, 2024.

Selected Conference Presentations

- "Vimentin Regulates Spheroid Poroelasticity to Enable Traction-Mediated Invasion."
  Oral presentation, Biomedical Engineering Society (BMES) Annual Meeting, Orlando, FL, October 23, 2026.
- Meet the Faculty Candidate Forum
  Selected poster presenter, BMES Annual Meeting, Orlando, FL, October 21, 2026.
- "Intercellular flow dominates the poroelasticity of the multicellular tissues."
  Poster presentation, BMES Annual Meeting, San Diego, CA, October 8–12, 2025.

TEACHING EXPERIENCE

Massachusetts Institute of Technology, Teaching Assistant

- Mechanics and Materials I (Undergraduate) — Spring 2024, 2025
- Fluid Mechanics (Graduate) — Fall 2020, 2021, 2023
- Mechanics of Solid Materials (Graduate) — Spring 2023, 2024
- Intermediate Heat and Mass Transfer (Undergraduate) — Fall 2021, 2022, 2023
- Introduction to Finite Element Methods (Undergraduate) — Spring 2021, 2023

Tsinghua University, Teaching Assistant

- Numerical Analysis (Graduate) — Fall 2017
- Mechanical Design Basis (Undergraduate) — Spring 2018

PROFESSIONAL SERVICE

- Session chair, "Cancer Mechanobiology I," BMES Annual Meeting, Orlando, FL, October 23, 2026.
- Journal reviewer: *Applied Physics Letters*, *Physics of Fluids*, *Biophysics Reviews*, *Materials Letters*, *Bioprinting*, *Biomedical Engineering Advances*, *Nano Energy*, and *Nanotechnology*.

PROFESSIONAL DEVELOPMENT

- **Generative Multiscale Materials Design: Physics, AI, Manufacturing**
  MIT Professional Education; short course, 3.0 CEUs. — Jun 2026

TECHNICAL SKILLS

- Software: MATLAB, COMSOL, Abaqus, SolidWorks, AutoCAD.
- Experimental: 3D printing, confocal and two-photon microscopy, optical tweezers.

[*CV compiled on 2026-11-02*]